

[Lecture Notebook] Dimension Reduction

MGS3701 Data Mining, Spring 2025

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Fri, 23 May 2025

Table 4.3

```
boston.housing.df <- read.csv("data/BostonHousing.csv", header = TRUE)
head(boston.housing.df, 9)
```

	CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO	LSTAT	MEDV
1	0.00632	18.0	2.31	0	0.538	6.575	65.2	4.0900	1	296	15.3	4.98	24.0
2	0.02731	0.0	7.07	0	0.469	6.421	78.9	4.9671	2	242	17.8	9.14	21.6
3	0.02729	0.0	7.07	0	0.469	7.185	61.1	4.9671	2	242	17.8	4.03	34.7
4	0.03237	0.0	2.18	0	0.458	6.998	45.8	6.0622	3	222	18.7	2.94	33.4
5	0.06905	0.0	2.18	0	0.458	7.147	54.2	6.0622	3	222	18.7	5.33	36.2
6	0.02985	0.0	2.18	0	0.458	6.430	58.7	6.0622	3	222	18.7	5.21	28.7
7	0.08829	12.5	7.87	0	0.524	6.012	66.6	5.5605	5	311	15.2	12.43	22.9
8	0.14455	12.5	7.87	0	0.524	6.172	96.1	5.9505	5	311	15.2	19.15	27.1
9	0.21124	12.5	7.87	0	0.524	5.631	100.0	6.0821	5	311	15.2	29.93	16.5

CAT..MEDV

1	0
2	0
3	1
4	1
5	1
6	0
7	0
8	0
9	0

```
summary(boston.housing.df)
```

	CRIM	ZN	INDUS	CHAS
26				
27	Min. : 0.00632	Min. : 0.00	Min. : 0.46	Min. :0.00000
28	1st Qu.: 0.08205	1st Qu.: 0.00	1st Qu.: 5.19	1st Qu.:0.00000
29	Median : 0.25651	Median : 0.00	Median : 9.69	Median :0.00000
30	Mean : 3.61352	Mean : 11.36	Mean :11.14	Mean :0.06917
31	3rd Qu.: 3.67708	3rd Qu.: 12.50	3rd Qu.:18.10	3rd Qu.:0.00000
32	Max. :88.97620	Max. :100.00	Max. :27.74	Max. :1.00000
33	NOX	RM	AGE	DIS
34	Min. :0.3850	Min. :3.561	Min. : 2.90	Min. : 1.130
35	1st Qu.:0.4490	1st Qu.:5.886	1st Qu.: 45.02	1st Qu.: 2.100
36	Median :0.5380	Median :6.208	Median : 77.50	Median : 3.207
37	Mean :0.5547	Mean :6.285	Mean : 68.57	Mean : 3.795
38	3rd Qu.:0.6240	3rd Qu.:6.623	3rd Qu.: 94.08	3rd Qu.: 5.188
39	Max. :0.8710	Max. :8.780	Max. :100.00	Max. :12.127
40	RAD	TAX	PTRATIO	LSTAT
41	Min. : 1.000	Min. :187.0	Min. :12.60	Min. : 1.73
42	1st Qu.: 4.000	1st Qu.:279.0	1st Qu.:17.40	1st Qu.: 6.95
43	Median : 5.000	Median :330.0	Median :19.05	Median :11.36
44	Mean : 9.549	Mean :408.2	Mean :18.46	Mean :12.65
45	3rd Qu.:24.000	3rd Qu.:666.0	3rd Qu.:20.20	3rd Qu.:16.95
46	Max. :24.000	Max. :711.0	Max. :22.00	Max. :37.97
47	MEDV	CAT..MEDV		
48	Min. : 5.00	Min. :0.000		
49	1st Qu.:17.02	1st Qu.:0.000		
50	Median :21.20	Median :0.000		
51	Mean :22.53	Mean :0.166		
52	3rd Qu.:25.00	3rd Qu.:0.000		
53	Max. :50.00	Max. :1.000		

54 **compute mean, standard dev., min, max, median, length, and missing values of CRIM**

```
mean(boston.housing.df$CRIM)
```

55 [1] 3.613524

```
sd(boston.housing.df$CRIM)
```

56 [1] 8.601545

```
min(boston.housing.df$CRIM)
```

57 [1] 0.00632

```
max(boston.housing.df$CRIM)
```

58 [1] 88.9762

```
median(boston.housing.df$CRIM)
```

59 [1] 0.25651

```
length(boston.housing.df$CRIM)
```

60 [1] 506

61 **find the number of missing values of variable CRIM**

```
sum(is.na(boston.housing.df$CRIM))
```

62 [1] 0

63 **compute mean, standard dev., min, max, median, length, and missing values for all**

```
data.frame(mean=sapply(boston.housing.df, mean),
            sd=sapply(boston.housing.df, sd),
            min=sapply(boston.housing.df, min),
            max=sapply(boston.housing.df, max),
            median=sapply(boston.housing.df, median),
            length=sapply(boston.housing.df, length),
            miss.val=sapply(boston.housing.df, function(x)
                           sum(length(which(is.na(x))))))
```

	mean	sd	min	max	median	length	miss.val
64 CRIM	3.61352356	8.6015451	0.00632	88.9762	0.25651	506	0
65 ZN	11.36363636	23.3224530	0.00000	100.0000	0.00000	506	0
66 INDUS	11.13677866	6.8603529	0.46000	27.7400	9.69000	506	0
67 CHAS	0.06916996	0.2539940	0.00000	1.0000	0.00000	506	0
68 NOX	0.55469506	0.1158777	0.38500	0.8710	0.53800	506	0
69 RM	6.28463439	0.7026171	3.56100	8.7800	6.20850	506	0

71	AGE	68.57490119	28.1488614	2.90000	100.0000	77.50000	506	0
72	DIS	3.79504269	2.1057101	1.12960	12.1265	3.20745	506	0
73	RAD	9.54940711	8.7072594	1.00000	24.0000	5.00000	506	0
74	TAX	408.23715415	168.5371161	187.00000	711.0000	330.00000	506	0
75	PTRATIO	18.45553360	2.1649455	12.60000	22.0000	19.05000	506	0
76	LSTAT	12.65306324	7.1410615	1.73000	37.9700	11.36000	506	0
77	MEDV	22.53280632	9.1971041	5.00000	50.0000	21.20000	506	0
78	CAT..MEDV	0.16600791	0.3724560	0.00000	1.0000	0.00000	506	0

Table 4.4

```
round(cor(boston.housing.df),2)
```

80		CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO
81	CRIM	1.00	-0.20	0.41	-0.06	0.42	-0.22	0.35	-0.38	0.63	0.58	0.29
82	ZN	-0.20	1.00	-0.53	-0.04	-0.52	0.31	-0.57	0.66	-0.31	-0.31	-0.39
83	INDUS	0.41	-0.53	1.00	0.06	0.76	-0.39	0.64	-0.71	0.60	0.72	0.38
84	CHAS	-0.06	-0.04	0.06	1.00	0.09	0.09	0.09	-0.10	-0.01	-0.04	-0.12
85	NOX	0.42	-0.52	0.76	0.09	1.00	-0.30	0.73	-0.77	0.61	0.67	0.19
86	RM	-0.22	0.31	-0.39	0.09	-0.30	1.00	-0.24	0.21	-0.21	-0.29	-0.36
87	AGE	0.35	-0.57	0.64	0.09	0.73	-0.24	1.00	-0.75	0.46	0.51	0.26
88	DIS	-0.38	0.66	-0.71	-0.10	-0.77	0.21	-0.75	1.00	-0.49	-0.53	-0.23
89	RAD	0.63	-0.31	0.60	-0.01	0.61	-0.21	0.46	-0.49	1.00	0.91	0.46
90	TAX	0.58	-0.31	0.72	-0.04	0.67	-0.29	0.51	-0.53	0.91	1.00	0.46
91	PTRATIO	0.29	-0.39	0.38	-0.12	0.19	-0.36	0.26	-0.23	0.46	0.46	1.00
92	LSTAT	0.46	-0.41	0.60	-0.05	0.59	-0.61	0.60	-0.50	0.49	0.54	0.37
93	MEDV	-0.39	0.36	-0.48	0.18	-0.43	0.70	-0.38	0.25	-0.38	-0.47	-0.51
94	CAT..MEDV	-0.15	0.37	-0.37	0.11	-0.23	0.64	-0.19	0.12	-0.20	-0.27	-0.44
95		LSTAT	MEDV	CAT..MEDV								
96	CRIM	0.46	-0.39		-0.15							
97	ZN	-0.41	0.36		0.37							
98	INDUS	0.60	-0.48		-0.37							
99	CHAS	-0.05	0.18		0.11							
100	NOX	0.59	-0.43		-0.23							
101	RM	-0.61	0.70		0.64							
102	AGE	0.60	-0.38		-0.19							
103	DIS	-0.50	0.25		0.12							
104	RAD	0.49	-0.38		-0.20							
105	TAX	0.54	-0.47		-0.27							
106	PTRATIO	0.37	-0.51		-0.44							
107	LSTAT	1.00	-0.74		-0.47							
108	MEDV	-0.74	1.00		0.79							
109	CAT..MEDV	-0.47	0.79		1.00							

110 **Table 4.5**

```
table(boston.housing.df$CHAS)
```

```
111
112    0    1
113 471   35
```

114 **Table 4.6**

```
# create bins of size 1
boston.housing.df$RM.bin <- .bincode(boston.housing.df$RM, c(1:9))

# compute the average of MEDV by (binned) RM and CHAS
# in aggregate() use the argument by= to define the list of aggregating variables,
# and FUN= as an aggregating function.
aggregate(boston.housing.df$MEDV, by=list(RM=boston.housing.df$RM.bin,
                                          CHAS=boston.housing.df$CHAS), FUN=mean)
```

```
115    RM CHAS      x
116  1   3    0 25.30000
117  2   4    0 15.40714
118  3   5    0 17.20000
119  4   6    0 21.76917
120  5   7    0 35.96444
121  6   8    0 45.70000
122  7   5    1 22.21818
123  8   6    1 25.91875
124  9   7    1 44.06667
125 10   8    1 35.95000
```

126 **Table 4.7**

```
# use install.packages("data/reshape") the first time the package is used
library(reshape)
boston.housing.df <- read.csv("data/BostonHousing.csv")
# create bins of size 1
```

```
boston.housing.df$RM.bin <- .bincode(boston.housing.df$RM, c(1:9))

# use melt() to stack a set of columns into a single column of data.
# stack MEDV values for each combination of (binned) RM and CHAS
#mlt <- melt(boston.housing.df, id=c("data/RM.bin", "CHAS"), measure=c("data/MEDV"))
#head(mlt, 5)

# use cast() to reshape data and generate pivot table
#cast(mlt, RM.bin ~ CHAS, subset=variable=="MEDV",
#      margins=c("data/grand_row", "grand_col"), mean)
```

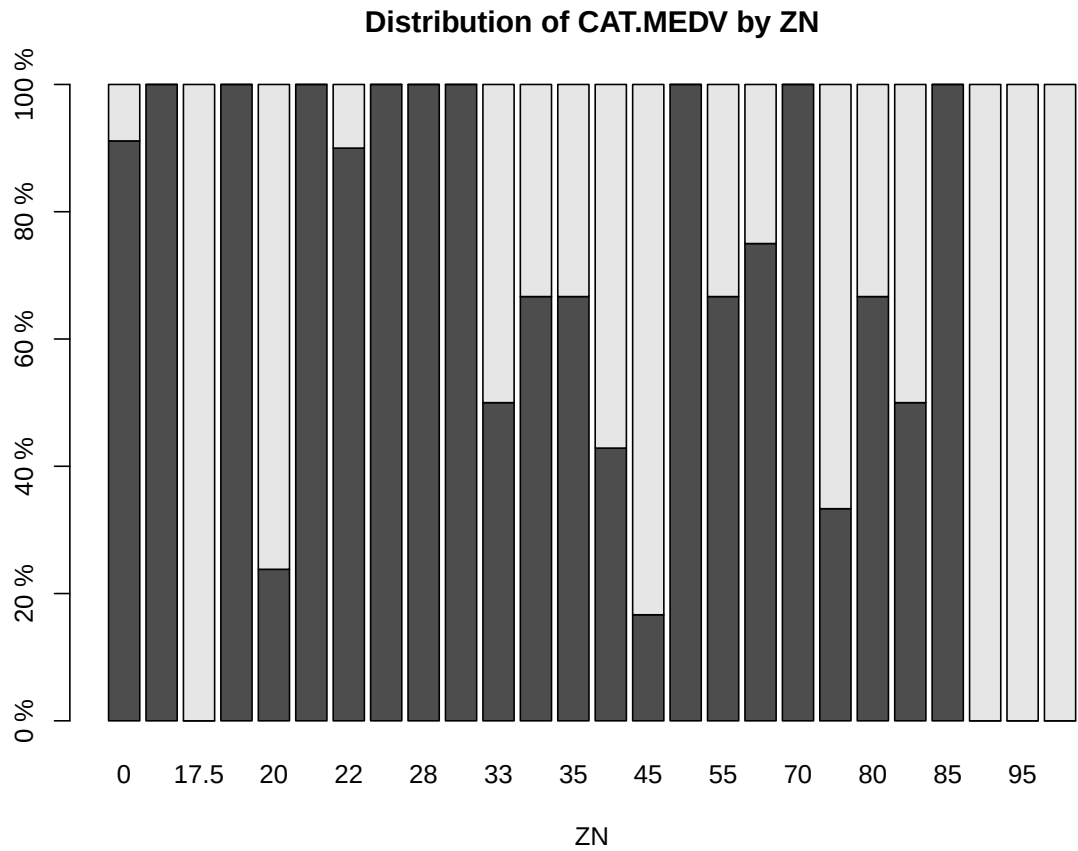
127 Figure 4.1

```
library(ggmap)
```

128 Loading required package: ggplot2

```
129 i Google's Terms of Service: <https://mapsplatform.google.com>
130   Stadia Maps' Terms of Service: <https://stadiamaps.com/terms-of-service>
131   OpenStreetMap's Tile Usage Policy: <https://operations.osmfoundation.org/policies/tiles>
132 i Please cite ggmap if you use it! Use `citation("ggmap")` for details.
```

```
tbl <- table(boston.housing.df$CAT..MEDV, boston.housing.df$ZN)
prop.tbl <- prop.table(tbl, margin=2)
barplot(prop.tbl, xlab="ZN", ylab="", yaxt="n", main="Distribution of CAT.MEDV by ZN")
axis(2, at=(seq(0,1, 0.2)), paste(seq(0,100,20), "%"))
```



133

134 **Table 4.10**

```
cereals.df <- read.csv("data/Cereals.csv")
# compute PCs on two dimensions
pcs <- prcomp(data.frame(cereals.df$calories, cereals.df$rating))
summary(pcs)
```

135 Importance of components:

136		PC1	PC2
137	Standard deviation	22.3165	8.8844
138	Proportion of Variance	0.8632	0.1368
139	Cumulative Proportion	0.8632	1.0000

```
pcs$rot
```

```
140          PC1      PC2
141 cereals.df.calories 0.8470535 0.5315077
142 cereals.df.rating  -0.5315077 0.8470535
```

```
scores <- pcs$x
head(scores, 5)
```

```
143          PC1      PC2
144 [1,] -44.921528  2.1971833
145 [2,]  15.725265 -0.3824165
146 [3,] -40.149935 -5.4072123
147 [4,] -75.310772 12.9991256
148 [5,]   7.041508 -5.3576857
```

Table 4.11

```
pcs <- prcomp(na.omit(cereals.df[, -c(1:3)]))
summary(pcs)
```

```
150 Importance of components:
151          PC1      PC2      PC3      PC4      PC5      PC6
152 Standard deviation  83.7641 70.9143 22.64375 19.18148 8.42323 2.09167
153 Proportion of Variance 0.5395 0.3867 0.03943 0.02829 0.00546 0.00034
154 Cumulative Proportion 0.5395 0.9262 0.96560 0.99389 0.99935 0.99968
155          PC7      PC8      PC9     PC10     PC11     PC12     PC13
156 Standard deviation  1.69942 0.77963 0.65783 0.37043 0.1864 0.06302 5.334e-08
157 Proportion of Variance 0.00022 0.00005 0.00003 0.00001 0.0000 0.00000 0.000e+00
158 Cumulative Proportion 0.99991 0.99995 0.99999 1.00000 1.0000 1.00000 1.000e+00
```

Table 4.12

```
pcs.cor <- prcomp(na.omit(cereals.df[, -c(1:3)]), scale. = T)
summary(pcs.cor)
```

160	Importance of components:							
161		PC1	PC2	PC3	PC4	PC5	PC6	PC7
162	Standard deviation	1.9062	1.7743	1.3818	1.00969	0.9947	0.84974	0.81946
163	Proportion of Variance	0.2795	0.2422	0.1469	0.07842	0.0761	0.05554	0.05166
164	Cumulative Proportion	0.2795	0.5217	0.6685	0.74696	0.8231	0.87861	0.93026
165		PC8	PC9	PC10	PC11	PC12	PC13	
166	Standard deviation	0.64515	0.56192	0.30301	0.25194	0.13897	1.499e-08	
167	Proportion of Variance	0.03202	0.02429	0.00706	0.00488	0.00149	0.000e+00	
168	Cumulative Proportion	0.96228	0.98657	0.99363	0.99851	1.00000	1.000e+00	